

LAN-BASED ONLINE EXAM

■ RAHUL BAGAI

Using this program, examinations can be conducted online for students of various subjects in schools and colleges. The examination result along with solutions to the questions is displayed at the end of the exam.

The questions are randomly generated from a bank of questions stored in a database maintained by the administrator. These are multiple-choice with four possible answers. Each question is worth one mark. The student has to select one subject at a time, so the questions will be on that topic only. For example, a subject may have many chapters and multiple questions related to that chapter are stored in the database. From here, only 15 questions from selected chapter are generated randomly for the test. The software has the facility to store many chapters under one particular subject. The exam paper will be made up of some questions taken from every topic/chapter.

The administrator can change this content by adding new topics and changing the number of questions for each topic. He can add any number of questions and topics.

The student can answer each question by clicking a radio button, so from four possible answers only one answer is selected per question. After answering all the questions, the student needs to click a button at the bottom. All the answers by the student are stored in the PC and the marks scored displayed.

Software program

The program is developed using J2SE software development kit (JDK1.6) and MS Access application on a Windows XP machine. It is designed on a client-server architecture which operates only when the server provides services

to the client. The question database is stored at the server end. The client accesses the database from the server machine and displays the contents on its output screen.

The project includes three Java source code files: ServerOnline.java,



Fig. 1: First page on the client PC

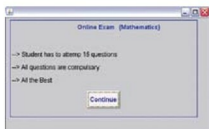


Fig. 2: Instructions for the students

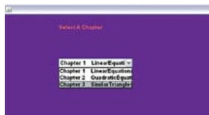


Fig. 3: Screenshot to select a chapter

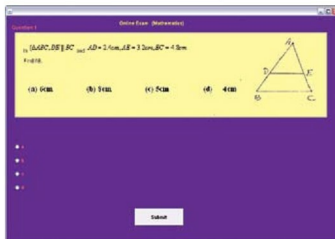


Fig. 4: Screenshot of the online question



ConnectDatabase1.java and ClientOnline.java. There is an Online Exams.mdb Microsoft Access database file which contains the list of questions and answers. The detailed questions are saved in GIF (graphic interchange format). The 'Questions and Answers' folder contains the images of the multiple-choice questions. There are also some other images and MP3 files included for the project. All the image and MP3 files are linked to the MS Access database through configurations as explained under the testing steps.

Operation of the program

1. As soon as the client program is started, a Java applet page as shown in Fig. 1 loads into its output screen. Here the student can select his class (10th or 12th) and the subject (physics, chemistry, biology or mathematics) for which he wants to give the exam. In the database for this project, we have included sample questions on biology and mathematics only. As soon as a radio button corresponding to a

particular subject is selected using mouse, the client machine needs to connect to the database stored in the server machine. So click 'Connect to Database' button to start the test.

2. Next, the instruction page is loaded. It displays some instructions to be